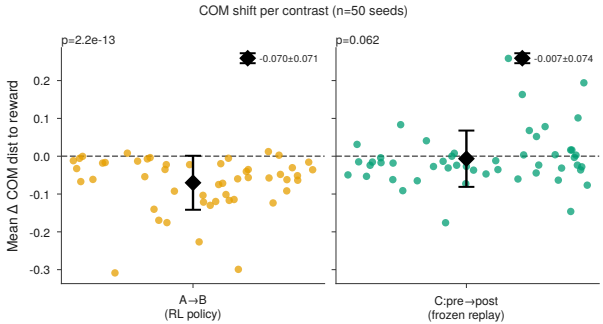
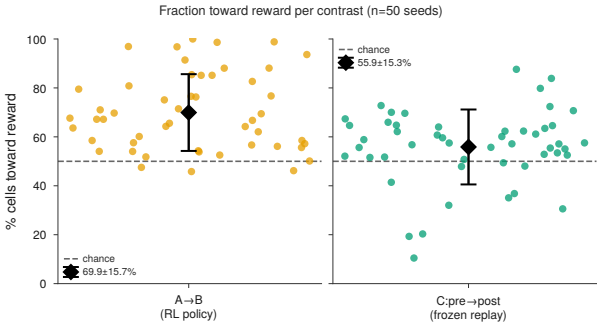
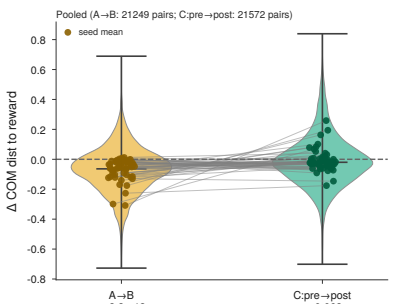
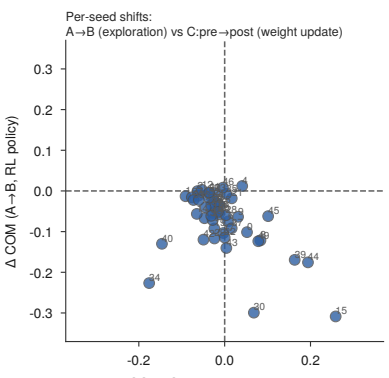
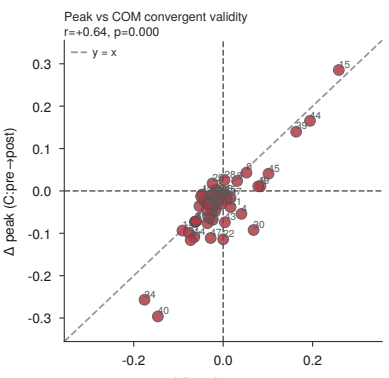

COM shift per contrast (n=50 seeds)
p=2.2e-13
-0.070±0.071
p=0.062
-0.007±0.074
Mean Δ COM dist to reward
A→B
(RL policy)
C:pre→post
(frozen replay)


Fraction toward reward per contrast (n=50 seeds)
% cells toward reward
chance
55.9±15.3%
chance
69.9±15.7%
A→B
(RL policy)
C:pre→post
(frozen replay)


Pooled (A→B: 21249 pairs; C:pre→post: 21572 pairs)
seed mean
Δ COM dist to reward
A→B
p=2.2e-13
C:pre→post
p=0.062


Per-seed shifts:
A→B (exploration) vs C:pre→post (weight update)
Δ COM (A→B, RL policy)
Δ COM (C:pre→post, frozen replay)


Peak vs COM convergent validity
r=+0.64, p=0.000
y = x
Δ peak (C:pre→post)
Δ COM (C:pre→post)